

BRAC UNIVERSITY

Department of Computer Science and Engineering

SET: A

Course: CSE423

Examination: Quiz 1

Semester: Spring 2025

Duration: 20 min

Marks: 10

Name:

ID:

1. A robot follows the path given $f(x, y) = \frac{x}{8} - \frac{y}{5} - 1 = 0$. If it moves along this line from the x-intercept to the y-intercept,
 - a) **Identify** the Zone of the above line. **3**
 - b) Using the Midpoint Line Drawing Algorithm, **calculate** the first 5(including the starting) pixels of that line segment. **5**
 - c) **[Bonus]** If an obstacle is placed at (4,3), check if the robot encounters it.
2. Write two differences between the Vector and Raster graphics. **2**